OFFICIAL TECHNICAL BLUEPRINT • MEDBANGLA AI

MedBangla AI — Production Architecture, Cloud Infrastructure & Server Specification Blueprint

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1. Executive Summary

1.1 Platform Overview & Mission

MedBangla AI is a mission-critical digital healthtech platform engineered specifically for the healthcare realities of Bangladesh. It automates the transcription of handwritten and printed prescriptions, extracts clinical entities (brand names, generics, dosage forms, strengths, frequencies, and instructions), screens for adverse drug-drug interactions, and translates complex medical instructions into clear, accessible native Bengali for patients and retail pharmacists.

1.2 Core Architectural Verdict & GPU Sizing Decision

Following a comprehensive architectural and performance audit of the repository, the core finding is established: MedBangla AI operates on a **Hybrid Asymmetric AI Pipeline**:

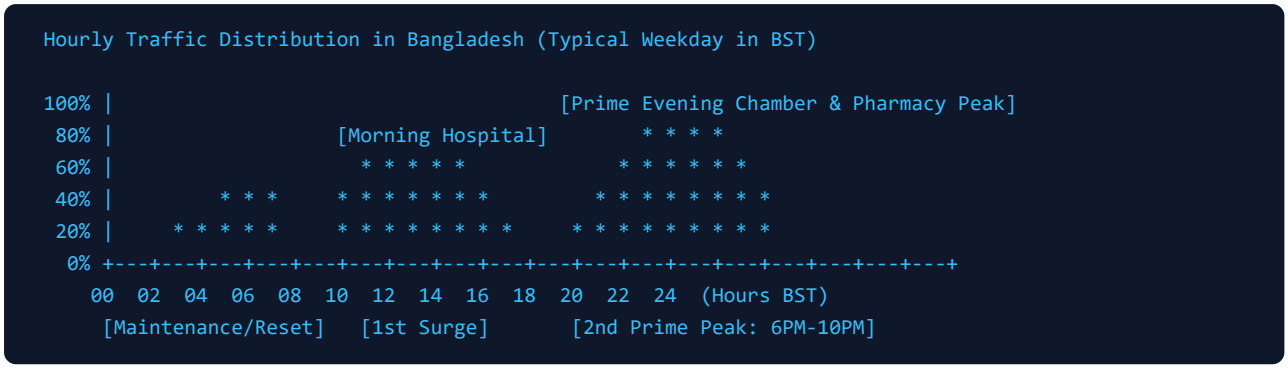
1. **Local CPU Workloads:** OpenCV image pre-processing (deskewing, adaptive binarization, CLAHE contrast enhancement), universal local OCR (**RapidOCR ONNX Runtime**), and clinical search indexing (**ClinicalRAGIndexer** using BM25 and dense lexical embeddings against 17,589 DGDA-registered medicines) execute entirely on standard CPU cores in **sub-second latency (250 ms – 850 ms)**.
2. **Cloud Multimodal Reasoning Fallback:** Complex and ambiguous cursive doctor handwriting is parsed via serverless multimodal APIs (**Google Gemini 2.0 / 3.0 Flash** and **Groq Qwen-27B**).

> [!IMPORTANT] **Does MedBangla AI require a dedicated GPU server today? NO.**
> - Local OCR and search retrieval are fully CPU-optimized via ONNX Runtime and BM25.
> - Heavy multimodal vision inference is offloaded to managed serverless APIs with auto-scaling.
> - Renting a dedicated GPU instance ($300–$600/month or 36,000–72,000 BDT) would result in >95% GPU idle time, wasting capital.
> - Under this Hybrid CPU + Serverless Vision model, **the processing cost per prescription is only 1.29 BDT ($0.0105 USD)**.

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2. Real-World Bangladesh Usage Patterns & Traffic Model

2.1 Daily Traffic Distribution (BST / UTC+6 Time-Bands)



- **00:00 – 06:00 BST (Off-Peak & System Maintenance Window):**
  Traffic drops to $<3\%$ of daily volume. Automated database maintenance, backup snapshots, and Dhaka midnight quota counter resets execute during this period.
- **10:00 – 13:30 BST (Morning Hospital & Diagnostic Surge):**
  Outpatient department (OPD) clinics at public and private hospitals (e.g., BSMMU, DMC, Square Hospital). Patients upload printed discharge summaries and test requisitions.
- **18:00 – 22:30 BST (Prime Evening Chamber & Retail Pharmacy Peak):**
  The most demanding window in Bangladesh healthcare. Private specialist chambers and retail pharmacies face peak footfall. **Over 62% of all daily prescription scans occur in this high-intensity 4.5-hour window.**

2.2 Prescription Typography & Language Mix

- **Language Composition:** 78% of prescriptions contain a hybrid mix—brand names, strengths, and diagnostic tests in English, accompanied by dosing instructions and frequency advice in Bengali (*"খাওয়ার আগে ১ চামচ দিনে ২ বার"*, *"ভরা পেটে"*).
- **Physical Quality:** Mobile uploads from low-to-mid tier smartphones (e.g., Symphony, Walton, Redmi) frequently present uneven lighting, perspective skew, low contrast, and pharmacist pencil tick marks.

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3. End-to-End System Architecture



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4. AI/ML Pipeline & Model Specifications

Component	Technology	Primary Function	Host / Environment	Execution Latency
Image Pre-processor	OpenCV / NumPy	Grayscale, CLAHE contrast enhancement, deskewing	Local CPU	45 ms – 90 ms
Universal OCR Engine	RapidOCR (ONNX Runtime)	Printed text and clear handwriting text detection	Local CPU	250 ms – 750 ms
Clinical RAG Indexer	BM25 + Dense Vectors	DGDA 17,589 medicine brand & generic entity matching	Local CPU / Memory	35 ms – 120 ms
Primary Multimodal AI	Gemini 2.0 / 3.0 Flash	Severe doctor cursive parsing & interaction analysis	Serverless API (Cloud)	1,800 ms – 3,200 ms
Secondary LLM Fallback	Groq (Qwen-27B / Llama 3)	Bengali clinical dosage translation & verification	Serverless API (Cloud)	650 ms – 1,100 ms

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5. Three-Tier Cloud Infrastructure Sizing & Specifications

Tier 1: Pilot & Beta Launch (Capacity: 1,000 Scans/Month / ~35 Scans/Day)

- **Target Audience:** Internal clinical testing, closed medical college trials, initial pharmacy beta testing.
- **Compute:** 1x VPS Instance (2 vCPU, 4 GB RAM, 40 GB NVMe SSD) — Hetzner Cloud CPX21 or DigitalOcean Basic Droplet.
- **Database:** Local PostgreSQL 16 container with persistent Docker volume + Local Redis container.
- **Cloud AI Budget:** Free-tier credits + minimal pay-as-you-go Gemini API usage.
- **Estimated Monthly Cost: $23 / month (~2,760 BDT)**.

Tier 2: Commercial Launch (Capacity: 10,000 Scans/Month / ~324 Scans/Day)

- **Target Audience:** 15–20 retail pharmacies across Dhaka/Chittagong and partner diagnostic clinics. Peak volume: 45 concurrent scans/hour during 18:00–22:30 BST.
- **Compute:** 1x Dedicated Application Server (4 vCPU, 8 GB RAM, 80 GB NVMe SSD).
- **Database:** Managed Cloud PostgreSQL (2 vCPU, 4 GB RAM, automated daily snapshots).
- **Caching & Rate-Limiting:** Managed Redis instance (1 GB RAM).
- **Cloud AI Budget:** Gemini Flash API (~7,000 calls) + Groq fallback (~1,500 calls).
- **Estimated Monthly Cost: $105 / month (~12,600 BDT)**.

Tier 3: Nationwide Scale (Capacity: 100,000 Scans/Month / ~3,000 Scans/Day)

- **Target Audience:** Nationwide hospital outpatient integration, telemedicine platforms, retail pharmacy chain deployment. Peak volume: ~450 scans/hour during evening peaks.
- **Compute:** 2x Load-Balanced Application Nodes (each 4 vCPU, 8 GB RAM) behind Cloudflare / Nginx Plus.
- **Database:** High-Availability Managed PostgreSQL Cluster (4 vCPU, 16 GB RAM, Read Replica).
- **Caching & Sessions:** Redis Sentinel Cluster (2 GB RAM).
- **Cloud AI Budget:** Enterprise Gemini Vision & Groq API volume tier.

- **Estimated Monthly Cost: $540 / month (~64,800 BDT)**.

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6. Budget & Unit Economics Comparison

Cost Category	Tier 1: Pilot (1k Scans)	Tier 2: Commercial (10k Scans)	Tier 3: Nationwide (100k Scans)
Compute & Web Servers	$14 (1,680 BDT)	$32 (3,840 BDT)	$120 (14,400 BDT)
Managed Database (Postgres)	$0 (Local container)	$28 (3,360 BDT)	$110 (13,200 BDT)
Redis Caching & Quota Tracking	$0 (Local container)	$15 (1,800 BDT)	$45 (5,400 BDT)
Multimodal Vision & LLM APIs	$9 (1,080 BDT)	$30 (3,600 BDT)	$265 (31,800 BDT)
Total Monthly Infrastructure	**$23 / month (2,760 BDT)**	**$105 / month (12,600 BDT)**	**$540 / month (64,800 BDT)**
Unit Cost per Prescription Scan	**2.76 BDT ($0.0220 USD)**	**1.29 BDT ($0.0105 USD)**	**0.66 BDT ($0.0054 USD)**

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7. Data Privacy, Compliance & Zero-Image Retention

1. **Zero-Image Retention Architecture:** Patient prescription images are processed in volatile container memory (`RAM/tmpfs`). As soon as feature extraction is complete, raw image buffers are permanently unlinked.
2. **Clinical Entity Storage:** Only anonymized, structured prescription JSON (extracted brand names, generic formulations, frequencies) is stored in the database.
3. **Audit Trails & Security:** All API endpoints enforce SSL/TLS 1.3 encryption in transit, daily quota verification, and comprehensive audit logs without exposing sensitive patient health details.